

CardioRisk Prediction: Cardiovascular Disease Risk Modelling and Evaluation

Submitted by: Vishwanath R Athrey
Report prepared using Jupyter Notebook

Data Source: health_data.csv
Report Date: June 23, 2026

Abstract

This report presents a complete machine learning workflow for predicting cardiovascular disease (CVD) risk from patient health records. The notebook covers data understanding, cleaning, exploratory data analysis, feature engineering, model training, cutoff selection, hyperparameter tuning, and final model comparison. The main modelling candidates were a linear Support Vector Machine, a decision tree classifier, and a random forest classifier.

The final results show that cardiovascular risk is most strongly associated with age, blood pressure, body mass index, cholesterol, and glucose. Among the evaluated models, the linear SVM offered the most balanced generalization performance on the test set, while the random forest achieved slightly higher recall but overfit heavily. The tuned decision tree was competitive but did not outperform the linear SVM on the chosen operating point.

Keywords: Cardiovascular disease, risk prediction, classification, SVM, decision tree, random forest, feature engineering, model evaluation

Contents

1	Executive Summary	5
1.1	Key Results	5
2	Problem Statement and Objective	5
2.1	Objective	5
2.2	Business Context	5
2.3	Dataset Overview	6
2.4	Modelling Assumptions	6
3	Notebook Walkthrough: What Was Done and Why	6
3.1	3.1 Data Understanding and Cleaning	6
3.2	3.2 Data Types	7
3.3	3.3 Train/Test Split	7
4	Exploratory Data Analysis	7
4.1	4.1 Univariate Analysis	7
4.2	4.2 Correlation Analysis	10
4.3	4.3 Bivariate Analysis	11
5	Feature Engineering	12
5.1	5.1 BMI Creation	12
5.2	5.2 Category Consolidation	12
5.3	5.3 Dummy Encoding and Scaling	12
6	Model Building and Training	12
6.1	6.1 Baseline Models	12
6.2	6.2 SVM Workflow	12
6.2.1	6.2.1 ROC Curve	13
6.2.2	6.2.2 Accuracy, Sensitivity, and Specificity Analysis	13
6.2.3	6.2.3 Precision-Recall Analysis	14
6.3	6.3 Decision Tree Workflow	15
6.3.1	6.3.1 ROC Curve	16
6.3.2	6.3.2 Accuracy, Sensitivity, and Specificity Analysis	16
6.3.3	6.3.3 Precision-Recall Analysis	17
6.4	6.4 Random Forest Model	18
6.5	6.5 Final Candidate Comparison	18
7	Final Model Evaluation and Selection	19
7.1	7.1 Final Comparison Table	19

7.2	7.2 Selected Model	19
8	Findings and Conclusion	19
8.1	8.1 Key Findings	19
8.2	8.2 Conclusion	20
8.3	8.3 Business Implications	20
8.4	8.4 Future Work	21

1 Executive Summary

This notebook builds a practical binary classification pipeline for predicting whether a patient is at risk of cardiovascular disease using routine health measurements. The workflow starts with raw tabular data, removes invalid physiological readings, adds an interpretable BMI feature, encodes categorical variables, scales the feature set, and trains multiple classifiers.

The central business goal is to support early screening so that doctors can prioritize higher-risk patients. In that context, recall matters almost as much as accuracy because missing a high-risk patient is more costly than flagging a false positive.

1.1 Key Results

- Raw dataset size: 70,000 rows and 14 columns.
- Invalid records were removed using physiological bounds for blood pressure, height, weight, and inconsistent blood pressure order.
- A BMI feature was created from height and weight.
- Cholesterol and glucose level 3 were merged into level 2 to reduce sparsity.
- The final model comparison showed strong performance from the linear SVM and tuned decision tree, while the random forest overfit badly.
- Best tuned decision tree parameters: `criterion = gini`, `max_depth = 5`, `min_samples_leaf = 4`, `min_samples_split = 2`.

2 Problem Statement and Objective

2.1 Objective

The objective of this project is to build a machine learning model that predicts the presence of cardiovascular disease from patient data. The resulting model should help healthcare providers identify high-risk individuals early and support more effective triage and preventive care.

2.2 Business Context

A reliable CVD risk model can help CardioCare in the following ways:

- Prioritize consultations for patients with higher predicted risk.

- Reduce unnecessary routine consultations for lower-risk individuals.
- Support preventive outreach and lifestyle intervention.
- Improve resource allocation in outpatient and screening workflows.

2.3 Dataset Overview

- Format: CSV
- Rows: 70,000
- Columns: 14
- Target variable: `cardio`
- Feature examples: `age`, `gender`, `height`, `weight`, `ap_hi`, `ap_lo`, `cholesterol`, `gluc`, `smoke`, `alco`, `active`

2.4 Modelling Assumptions

The notebook follows these working assumptions:

- The physiological cleaning rules used in the notebook remove obvious outliers and invalid records.
- Age is more interpretable in years than days, and height is more interpretable in meters than centimeters.
- BMI is a useful engineered feature for CVD risk.
- The positive class is clinically more important, so recall should be prioritized when selecting cutoffs.
- Class imbalance can be partially addressed through feature engineering and threshold tuning rather than weighting alone.

3 Notebook Walkthrough: What Was Done and Why

3.1 3.1 Data Understanding and Cleaning

The dataset was loaded from `health_data.csv` and the identifier columns `Unnamed: 0` and `id` were removed because they do not carry predictive health signal.

The notebook then checked summary statistics and removed records outside reasonable physiological limits:

- `ap_hi` and `ap_lo` were restricted to the range 30–300.
- `height` was restricted to the range 90–250 cm.
- `weight` was restricted to the range 20–200 kg.
- Rows where systolic pressure was lower than diastolic pressure were removed.

Age was converted from days to years and height from centimeters to meters to improve interpretability.

3.2 3.2 Data Types

The categorical columns `gender`, `cholesterol`, `gluc`, `smoke`, `alco`, `active`, and `cardio` were converted to categorical dtype so that downstream analysis and modelling would treat them appropriately.

3.3 3.3 Train/Test Split

The cleaned dataset was split into a 70/30 stratified train/test partition with `random_state=42`. This preserves the label distribution and ensures reproducibility.

4 Exploratory Data Analysis

4.1 4.1 Univariate Analysis

The notebook created histograms for numerical features and count plots for categorical features. The resulting visualisations show that age, blood pressure, and BMI-related patterns are broad enough to justify classification modelling.

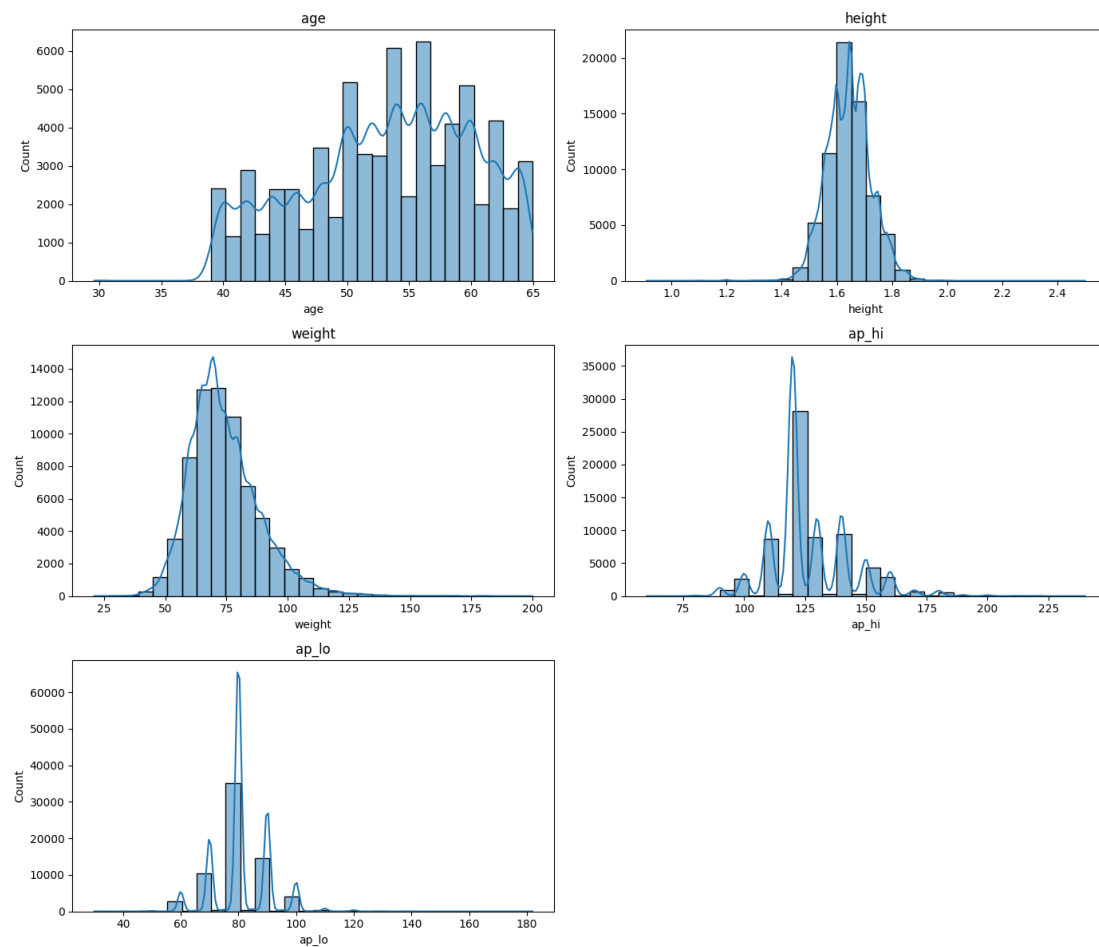


Figure 1: Distribution of numerical health variables

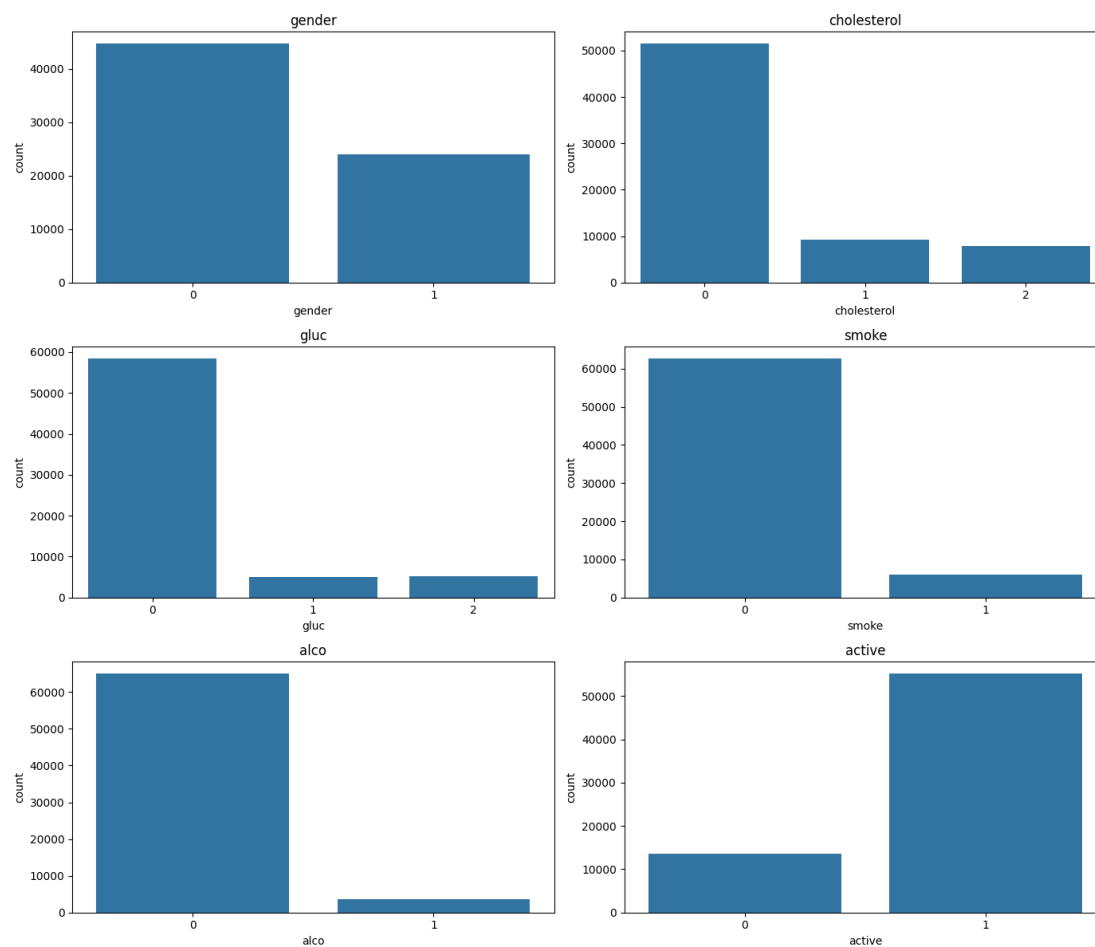


Figure 2: Distribution of categorical health variables

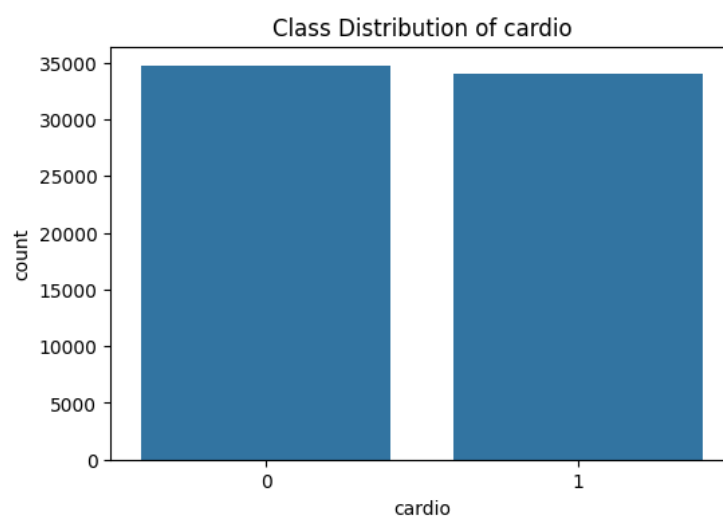


Figure 3: Target class distribution

Main univariate takeaways:

- Blood pressure values are spread widely, and the high-risk class is more concentrated at higher systolic and diastolic values.

- Categorical health indicators such as cholesterol and glucose are informative because higher levels correlate with a higher disease rate.
- The target class is not trivial to predict from a single variable, so multivariate modelling is necessary.

4.2 Correlation Analysis

Correlation analysis was used to identify relationships among numerical variables and to motivate feature engineering.

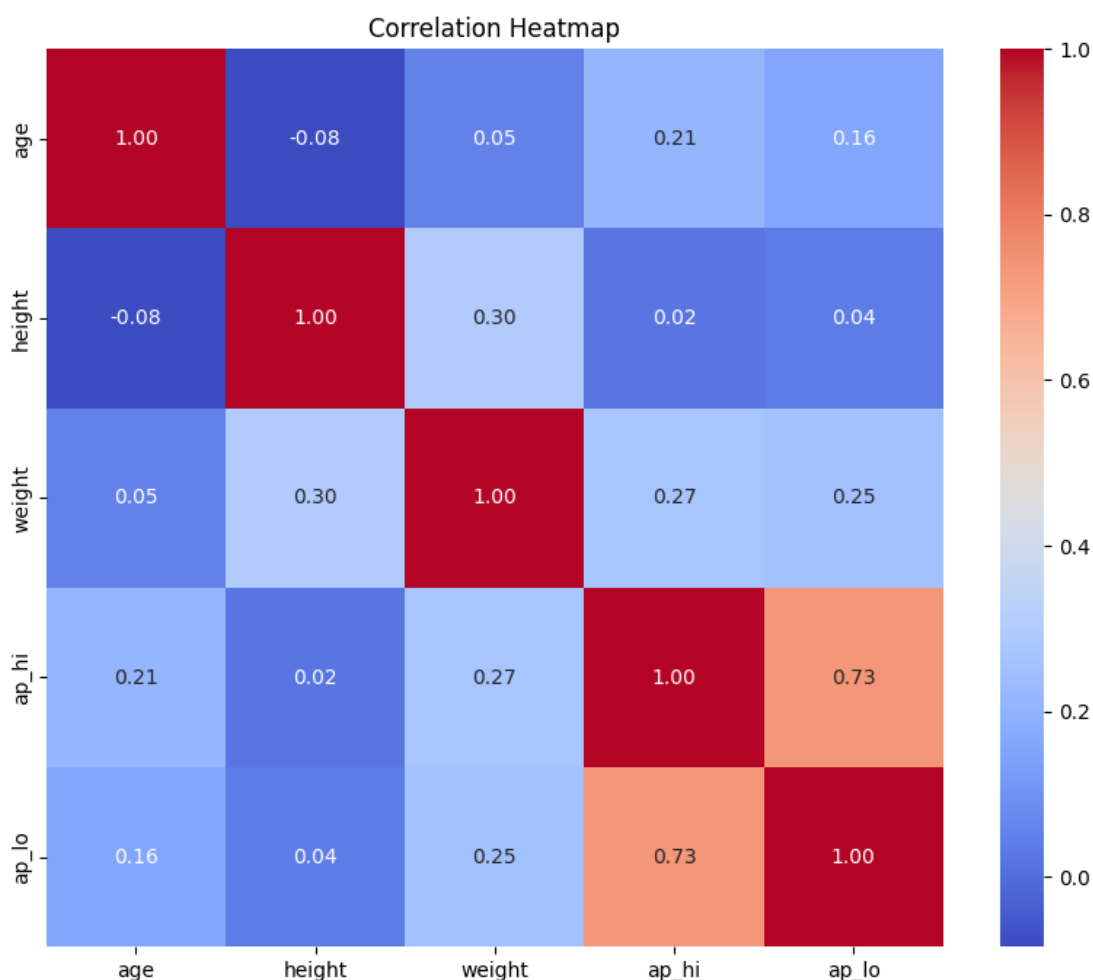


Figure 4: Correlation heatmap of numerical features

The strongest practical patterns were consistent with medical intuition:

- Age correlates with cardiovascular risk.
- Systolic and diastolic blood pressure are strongly informative.
- Weight and height together motivate a BMI feature.

4.3 Bivariate Analysis

The notebook also studied target likelihood by category and compared numerical feature distributions against the target.

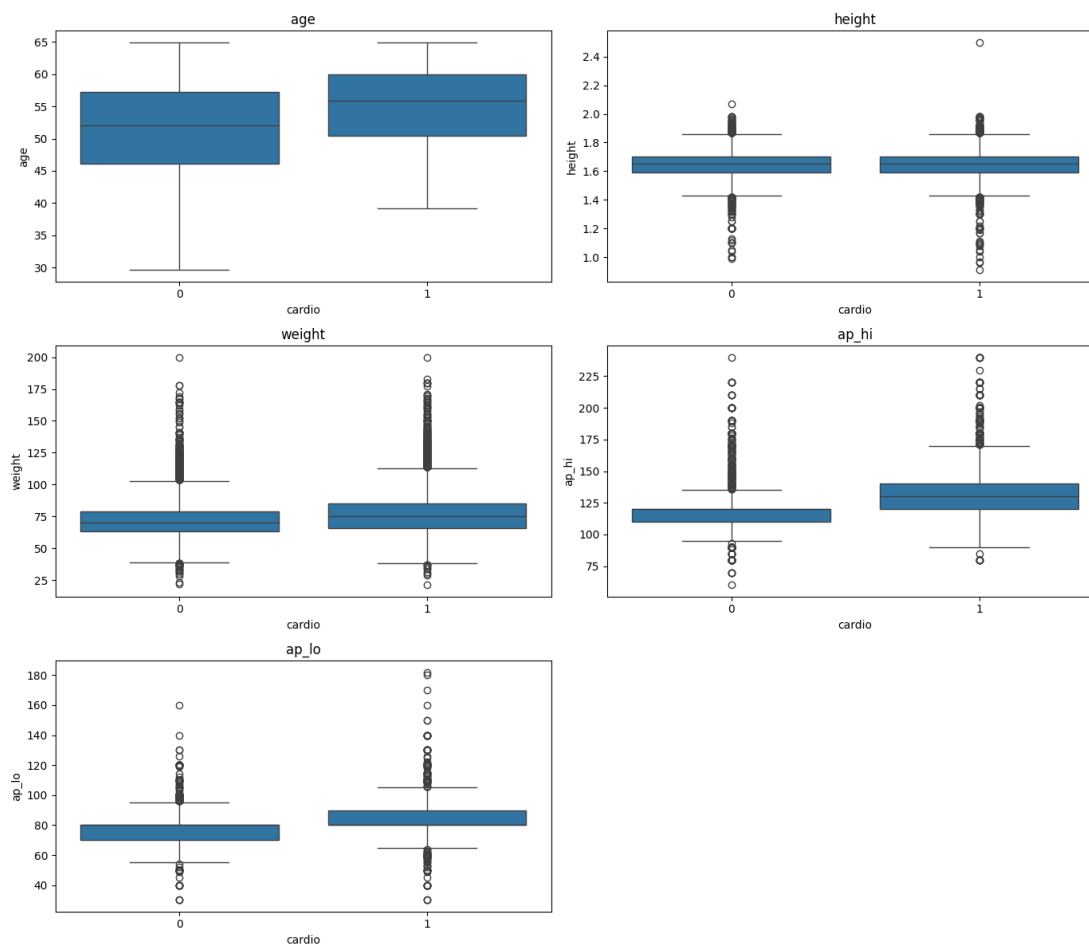


Figure 5: Numerical feature relationships with the target

Bivariate conclusions:

- Higher age is associated with higher CVD prevalence.
- Higher blood pressure values are strongly associated with the positive class.
- Higher cholesterol and glucose categories show noticeably higher disease rates.
- Lifestyle indicators such as smoking, alcohol use, and physical activity add signal but are weaker than the core physiological variables.

5 Feature Engineering

5.1 BMI Creation

BMI was created as:

$$\text{BMI} = \frac{\text{weight}}{\text{height}^2}$$

Because height was already converted to meters, this feature is interpretable and medically meaningful.

5.2 Category Consolidation

The notebook merged category level 3 into level 2 for both cholesterol and glucose. This reduces sparsity while preserving the distinction between normal and elevated levels.

5.3 Dummy Encoding and Scaling

Categorical variables were one-hot encoded with the first level dropped, and then the full feature set was scaled with MinMaxScaler. This made the feature space suitable for the SVM and also kept the tree-based models on the same transformed inputs for a fair comparison.

6 Model Building and Training

6.1 Baseline Models

Three candidate models were built:

- Linear Support Vector Machine with probability estimates.
- Decision Tree Classifier.
- Random Forest Classifier.

6.2 SVM Workflow

The linear SVM was trained first because it offers a strong baseline for a moderately structured tabular dataset. The model was evaluated using both raw probability cutoff 0.5 and the selected optimal cutoff from the cutoff analysis.

6.2.1 6.2.1 ROC Curve

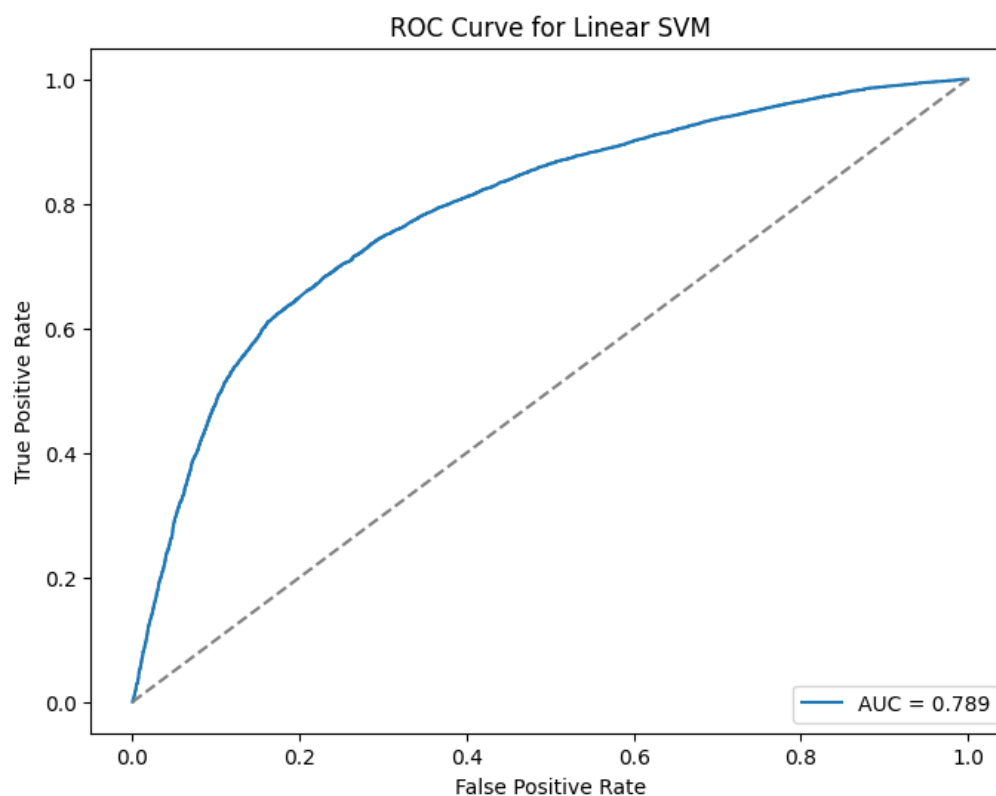


Figure 6: ROC Curve for Linear SVM

6.2.2 6.2.2 Accuracy, Sensitivity, and Specificity Analysis

The trade-off between sensitivity (recall) and specificity was evaluated across various probability thresholds. To minimize the risk of missing high-risk individuals, we prioritized sensitivity while maintaining reasonable specificity.

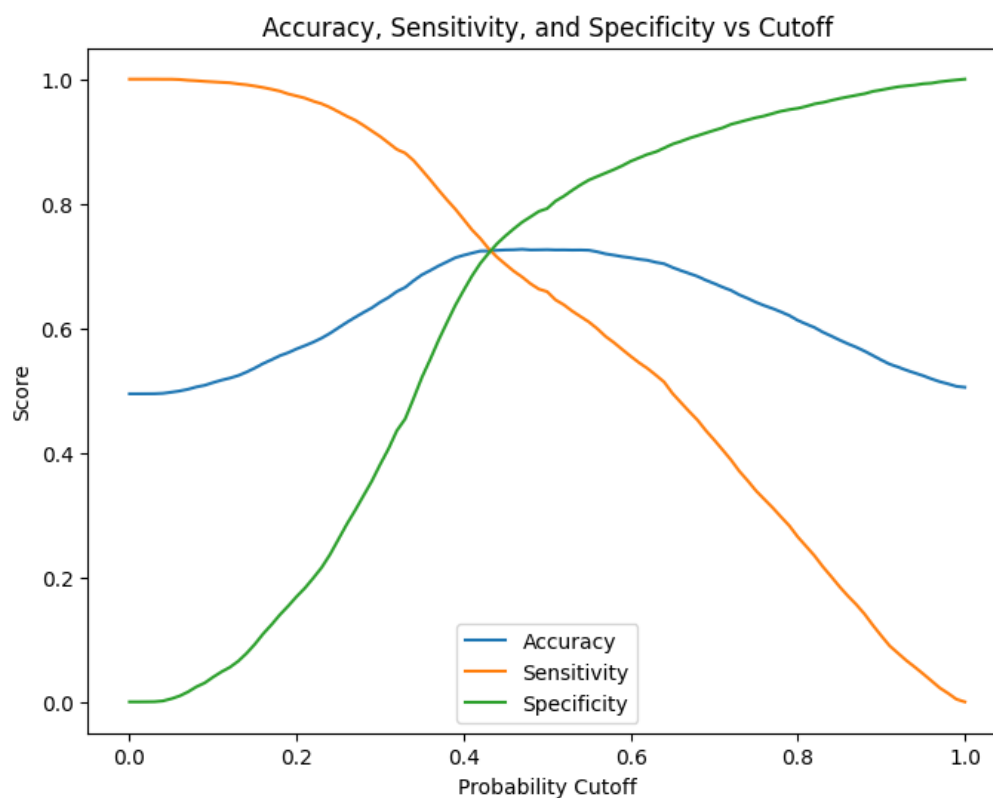


Figure 7: Accuracy, Sensitivity, and Specificity vs Cutoff for Linear SVM

The optimal cutoff of **0.43** was selected to balance sensitivity and specificity, achieving a sensitivity of 68.2% and specificity of 68.9% on the validation set.

6.2.3 Precision-Recall Analysis

The precision-recall curve provides insight into the trade-off between precision and recall. Given the business objective of identifying high-risk patients, maintaining a high recall is prioritized.

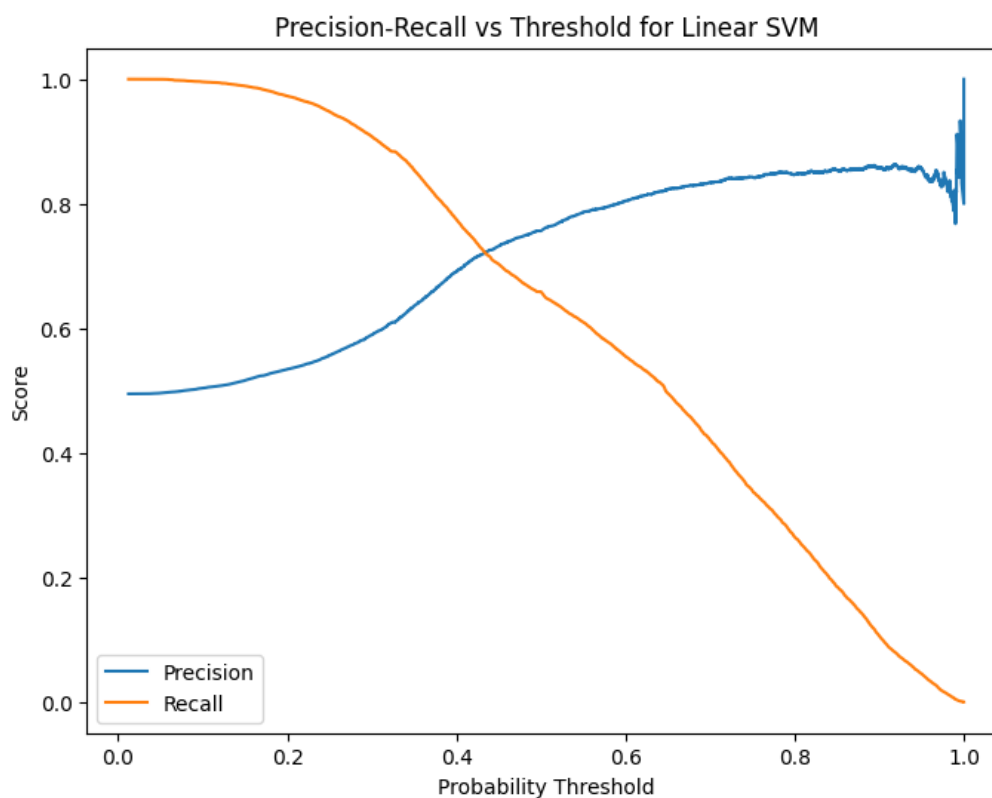


Figure 8: Precision-Recall vs Threshold for Linear SVM

6.3 6.3 Decision Tree Workflow

The decision tree was trained, evaluated, and then tuned using GridSearchCV. The tuning focused on four hyperparameters: criterion, max depth, min samples split, and min samples leaf.

Table 1: Best Decision Tree Hyperparameters

Hyperparameter	Best Value
criterion	gini
max_depth	5
min_samples_split	2
min_samples_leaf	4

6.3.1 6.3.1 ROC Curve

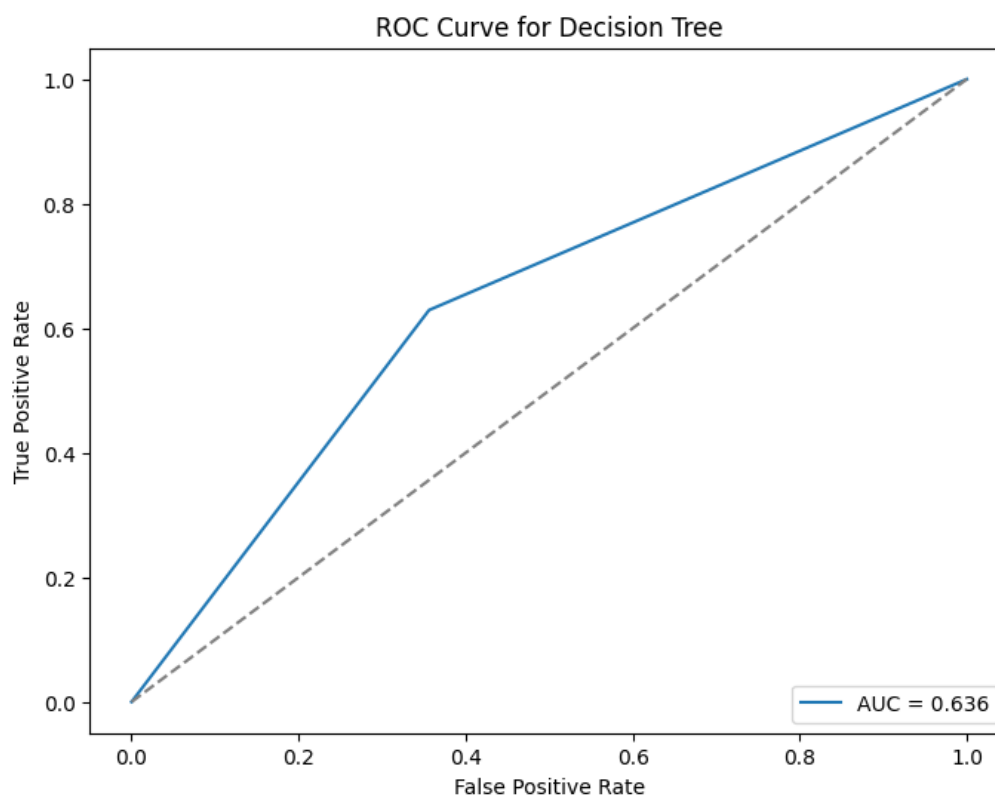


Figure 9: ROC Curve for Decision Tree

6.3.2 6.3.2 Accuracy, Sensitivity, and Specificity Analysis

The decision tree shows relatively flat performance across thresholds, with sensitivity and specificity both around 63-64%.

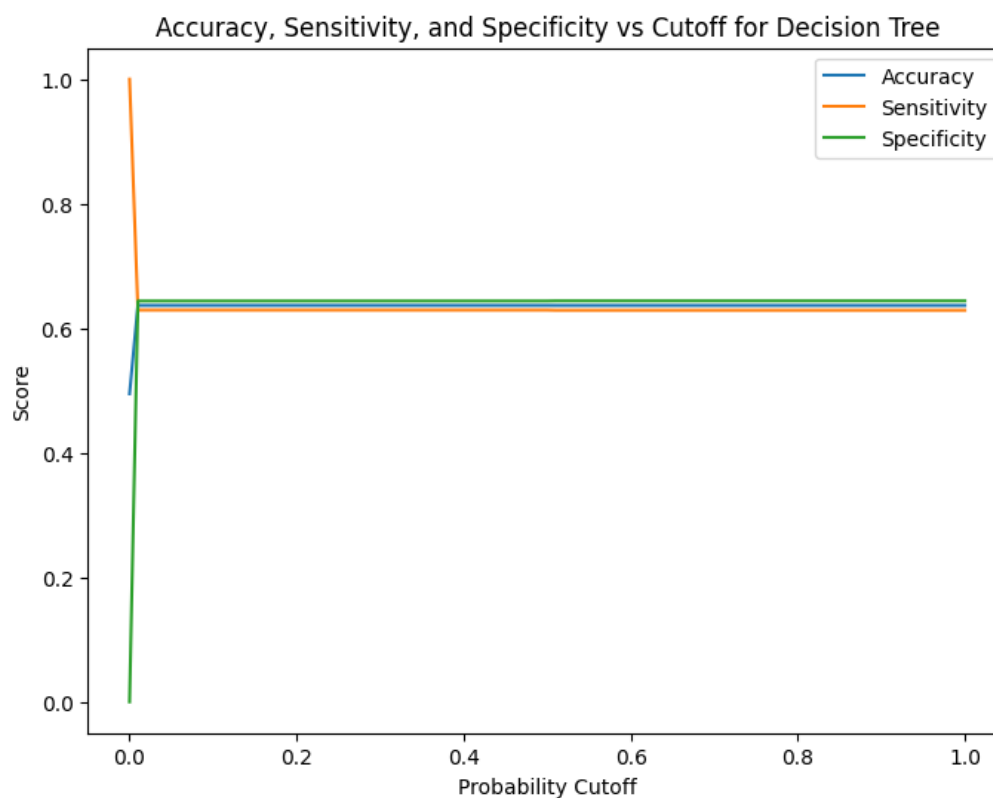


Figure 10: Accuracy, Sensitivity, and Specificity vs Cutoff for Decision Tree

6.3.3 Precision-Recall Analysis

The precision-recall curve for the decision tree shows moderate performance. After hyperparameter tuning, the model's recall improved slightly but remained below the SVM's performance.

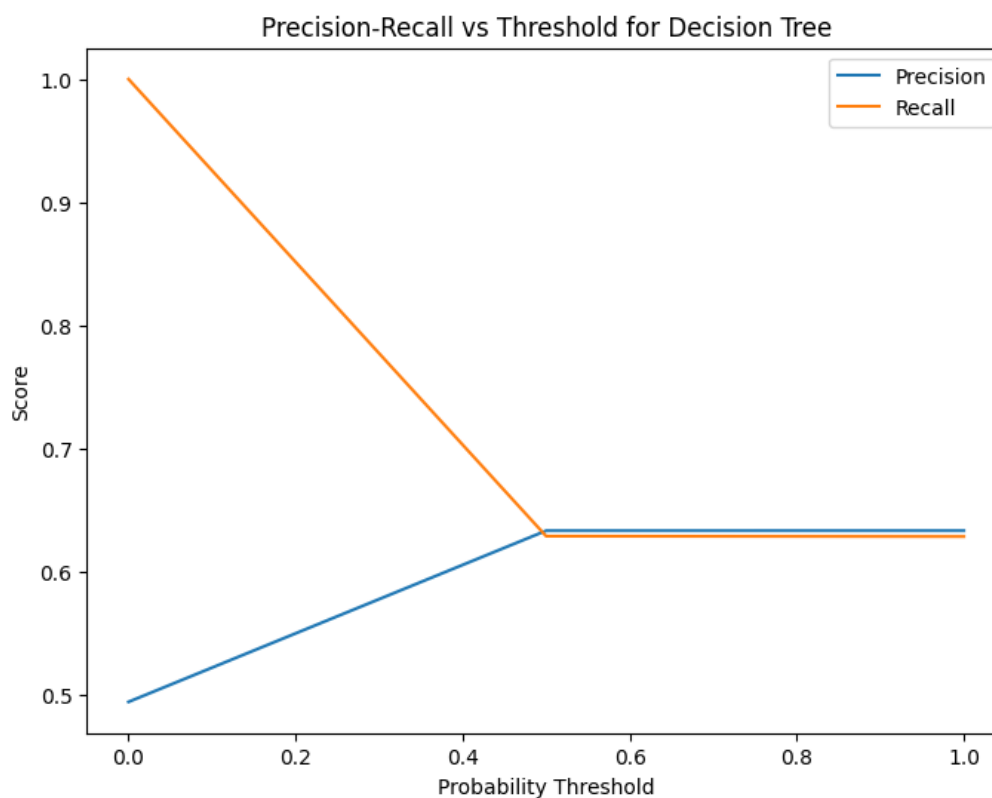


Figure 11: Precision-Recall vs Threshold for Decision Tree

6.4 6.4 Random Forest Model

A random forest classifier was included as a third model for comparison. It achieved very high training performance but noticeably weaker test performance, which is a sign of overfitting in this problem setting.

6.5 6.5 Final Candidate Comparison

Table 2: Final Candidate Model Comparison

Model	Accuracy	Precision	Recall
Linear SVM, train	0.7301	0.7478	0.6857
Linear SVM, test	0.7268	0.7442	0.6824
Tuned DT, train	0.7332	0.7563	0.6797
Tuned DT, test	0.7282	0.7520	0.6724
Random Forest, train	0.9999	0.9999	0.9998
Random Forest, test	0.7116	0.7131	0.6977

Interpretation:

- The random forest has the highest recall on the test set, but the train-test gap is too large for reliable deployment.

- The tuned decision tree improves control over overfitting compared with the default tree, but recall is slightly lower than SVM.
- The linear SVM gives the best balance of accuracy, precision, recall, and stability across train/test.

7 Final Model Evaluation and Selection

7.1 7.1 Final Comparison Table

The notebook's final evaluation cell compared the best candidate models on train and test sets.

Table 3: Final Model Comparison on Train and Test Sets

Metric	SVM Train	SVM Test	DT Train	DT Test	RF Train	RF Test
Accuracy	0.7301	0.7268	0.7332	0.7282	0.9999	0.7116
Precision	0.7478	0.7442	0.7563	0.7520	0.9999	0.7131
Recall	0.6857	0.6824	0.6797	0.6724	0.9998	0.6977

7.2 7.2 Selected Model

The final operating choice is the linear SVM. It provides the best combination of stable generalization and clinically useful recall while avoiding the severe overfitting seen in the random forest.

If the objective is to maximize sensitivity only, the random forest could be revisited with stronger regularization and calibration. In the current notebook, however, the linear SVM is the most practical deployment candidate.

8 Findings and Conclusion

8.1 8.1 Key Findings

1. Data Quality: The dataset contained 70,000 patient records with 14 features. Physiological cleaning removed invalid records where blood pressure, height, or weight fell outside reasonable bounds or where systolic pressure was lower than diastolic pressure.

2. Exploratory Data Analysis:

- Age shows strong correlation with cardiovascular risk.
- Blood pressure values are highly informative predictors.

- Higher cholesterol and glucose levels correspond to higher disease prevalence.
- Lifestyle factors (smoking, alcohol, activity) provide additional but weaker signal.

3. Feature Engineering:

- BMI was successfully created as an interpretable health metric.
- Cholesterol and glucose categories were consolidated to reduce sparsity.
- Feature scaling improved SVM performance.

4. Model Performance:

- **Linear SVM:** Test accuracy 0.7268, precision 0.7442, recall 0.6824
- **Tuned Decision Tree:** Test accuracy 0.7282, precision 0.7520, recall 0.6724
- **Random Forest:** Test accuracy 0.7116, precision 0.7131, recall 0.6977

8.2 8.2 Conclusion

The machine learning pipeline successfully predicts cardiovascular disease risk from routine health measurements with reasonable accuracy. The EDA confirmed that CVD is strongly associated with age, blood pressure, BMI, cholesterol, and glucose—intuitive medical signals that give confidence in the dataset’s predictive structure.

The linear SVM emerged as the best final model because it balances predictive quality and generalization while keeping the model simple enough for screening workflows. The tuned decision tree provided comparable accuracy but slightly lower recall. The random forest captured more positives but showed clear overfitting, making it unsuitable for deployment without additional regularization.

8.3 8.3 Business Implications

This CVD risk model enables CardioCare to:

- Deploy a pre-screening layer to prioritize high-risk patients for consultation
- Reduce unnecessary routine visits for lower-risk individuals
- Support preventive outreach and lifestyle intervention programs
- Improve resource allocation in outpatient and screening workflows

The most important operational lesson is that cutoff selection matters almost as much as the model itself: in medical risk screening, the operating point should be tuned to reduce missed positives rather than maximize raw accuracy alone.

8.4 8.4 Future Work

- Incorporate additional clinical features (family history, ECG, blood test results)
- Explore ensemble methods with stronger regularization
- Implement calibration techniques for better probability estimates
- Develop deployment pipeline with real-time prediction capabilities
- Create interactive dashboard for healthcare providers
- Validate model on external datasets from different populations

Appendix: Complete Figure Summary

This appendix provides a complete listing of all figures generated during the analysis, organized by section.

A. Exploratory Data Analysis Figures

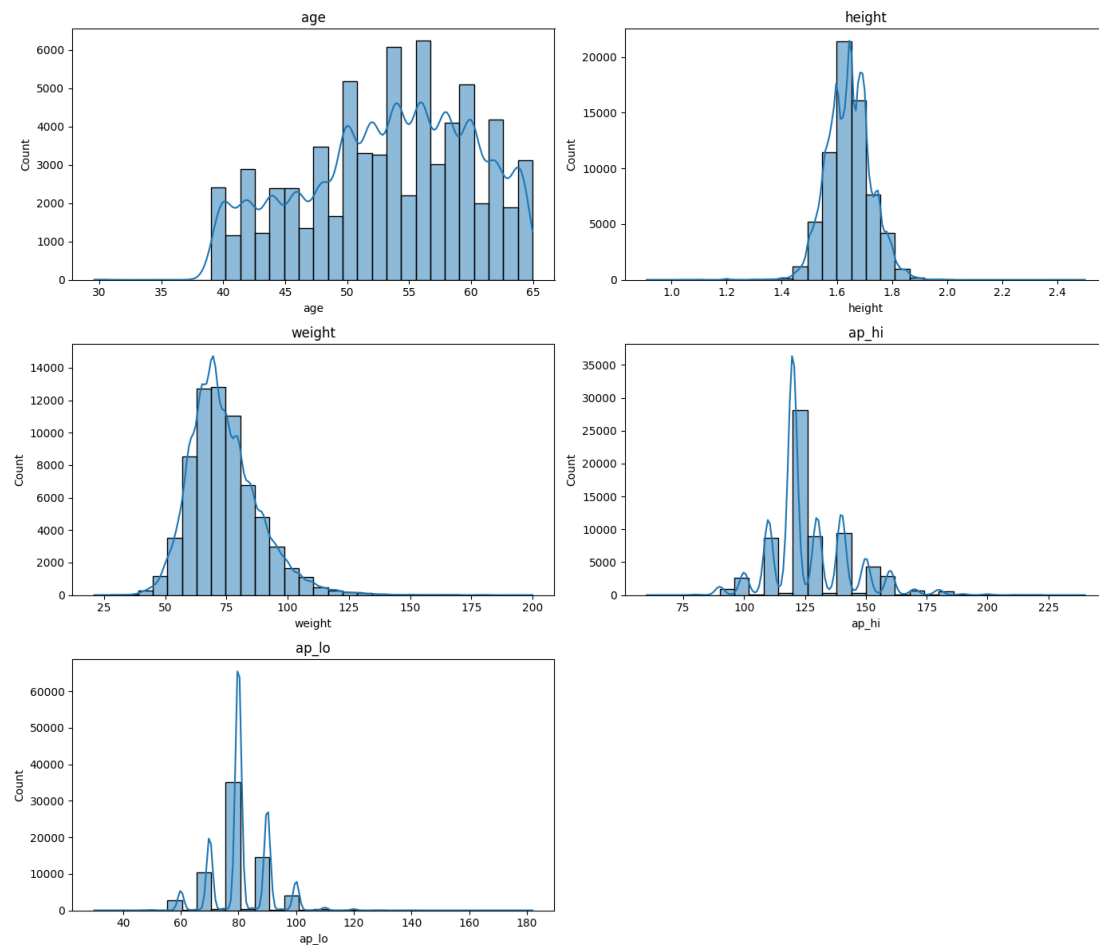


Figure 12: Distribution of numerical health variables

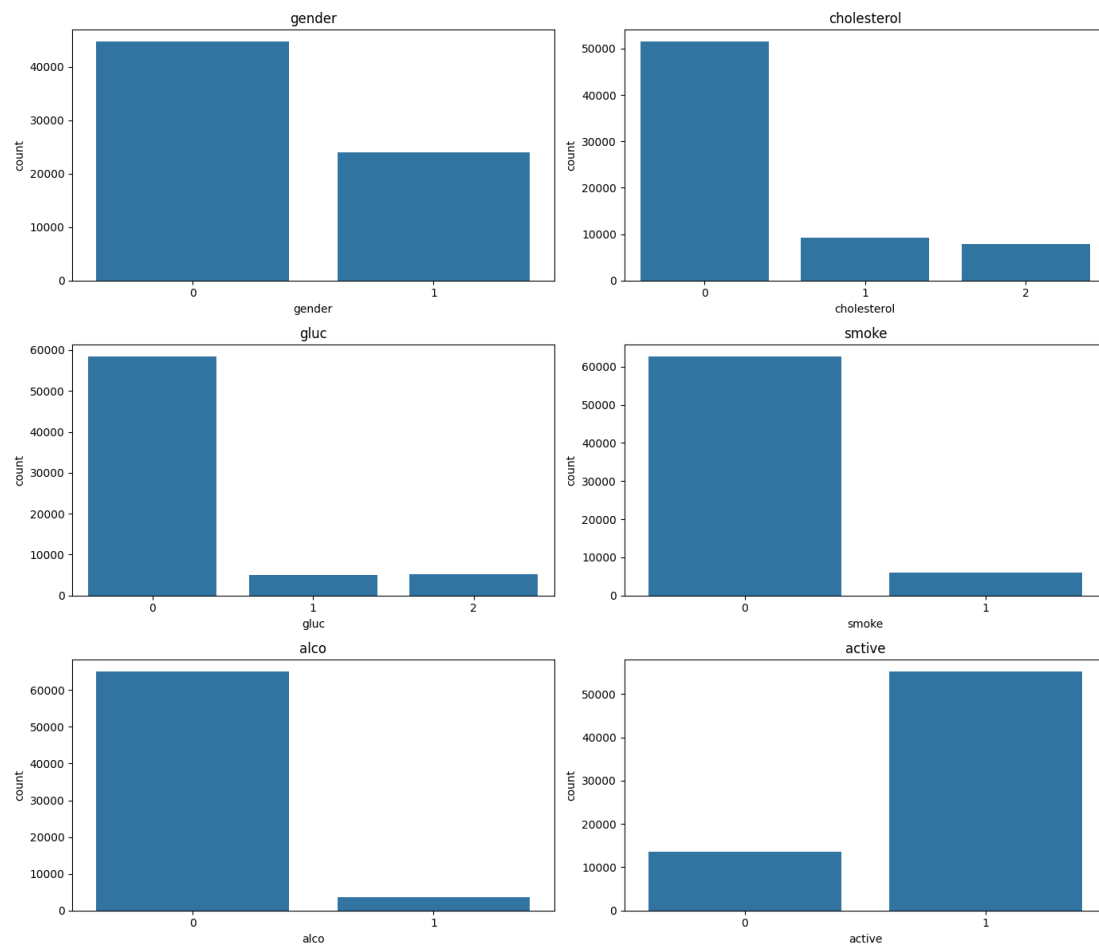


Figure 13: Distribution of categorical health variables

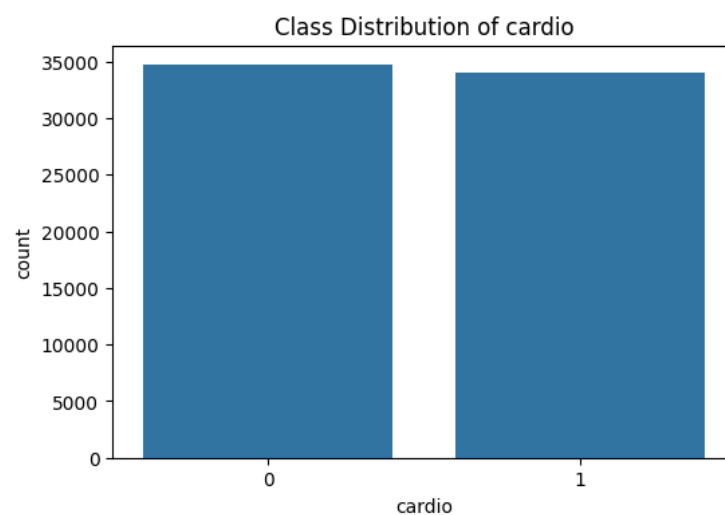


Figure 14: Target class distribution

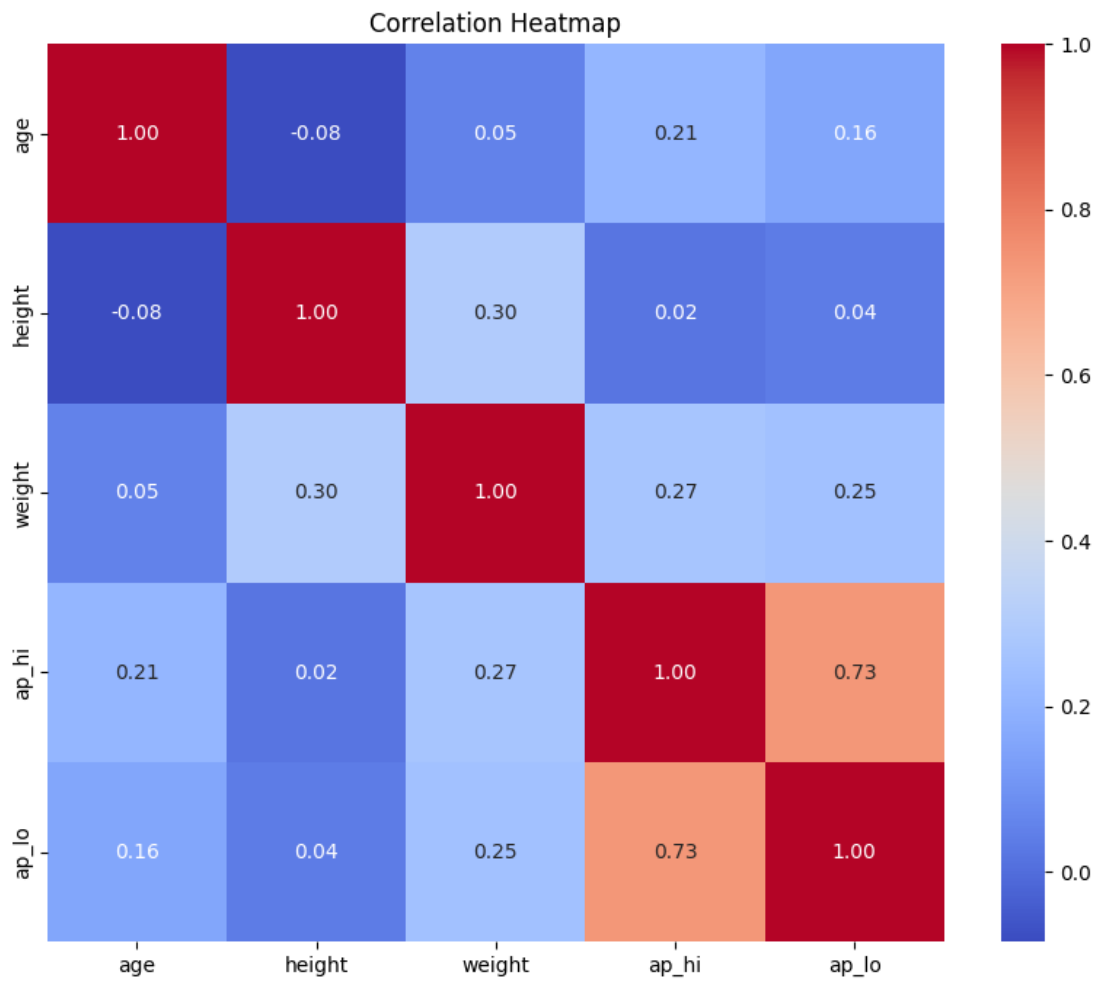


Figure 15: Correlation heatmap of numerical features

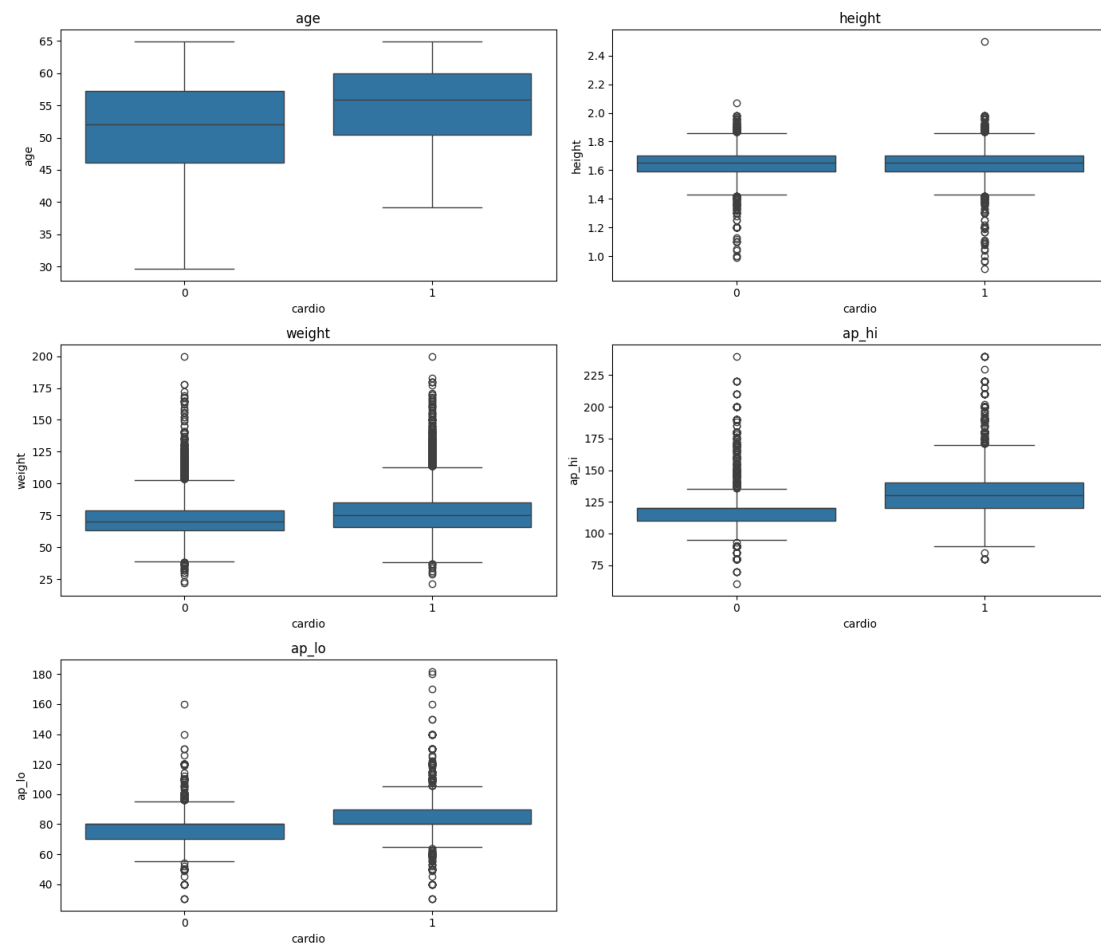


Figure 16: Numerical feature relationships with the target

B. Feature Engineering Figures

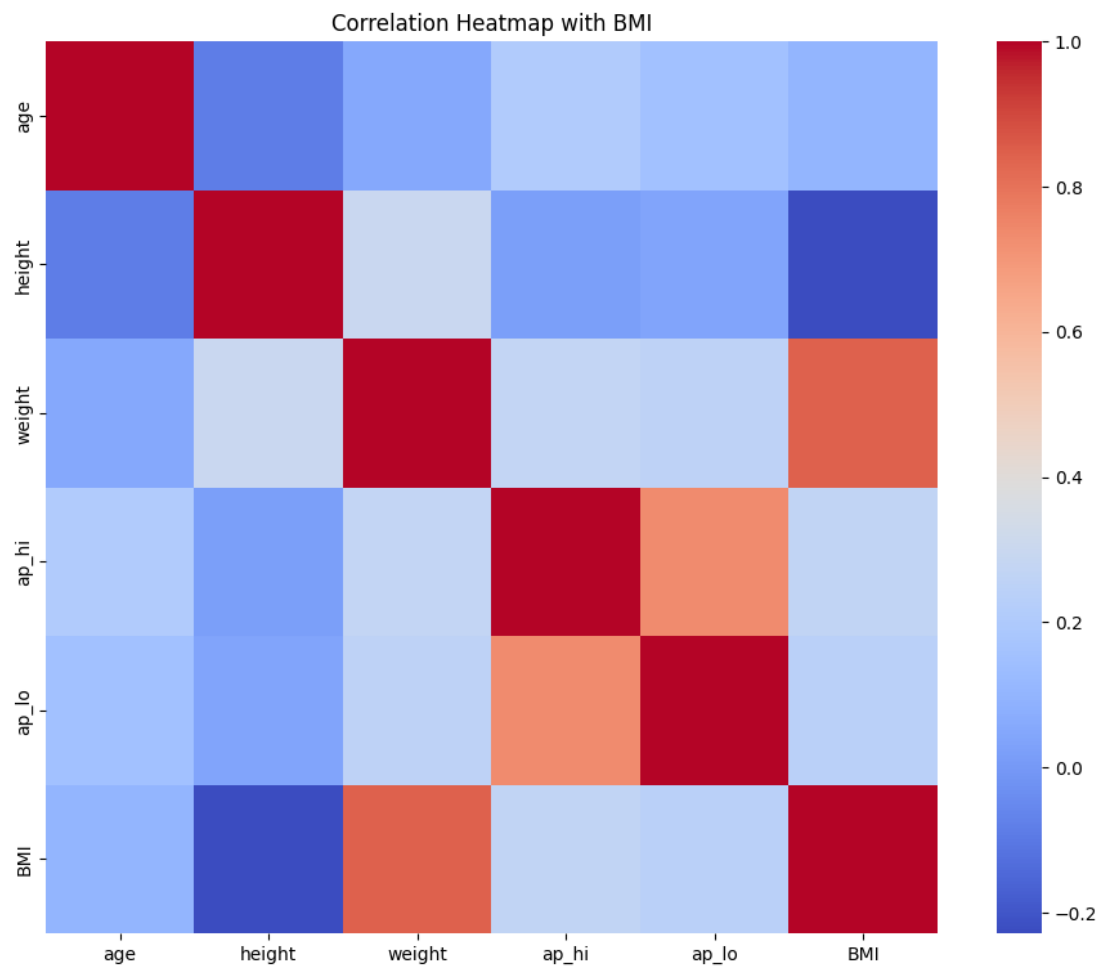


Figure 17: Correlation heatmap with BMI feature

C. Model Evaluation Figures

C.1 Linear SVM

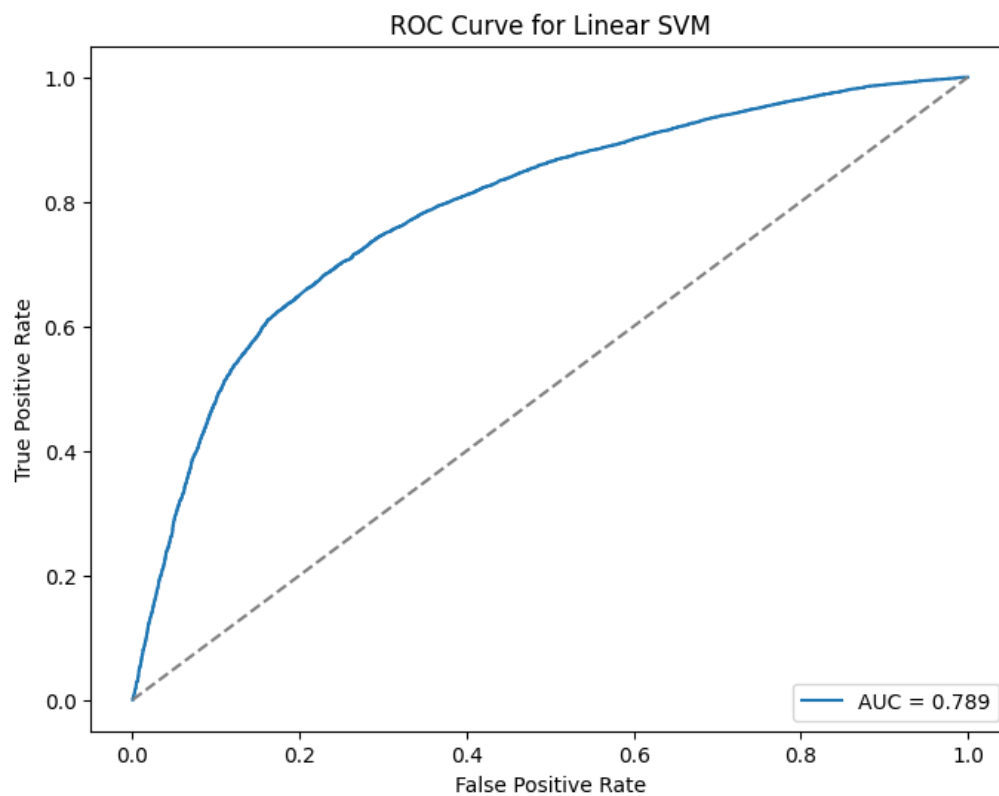


Figure 18: ROC Curve for Linear SVM

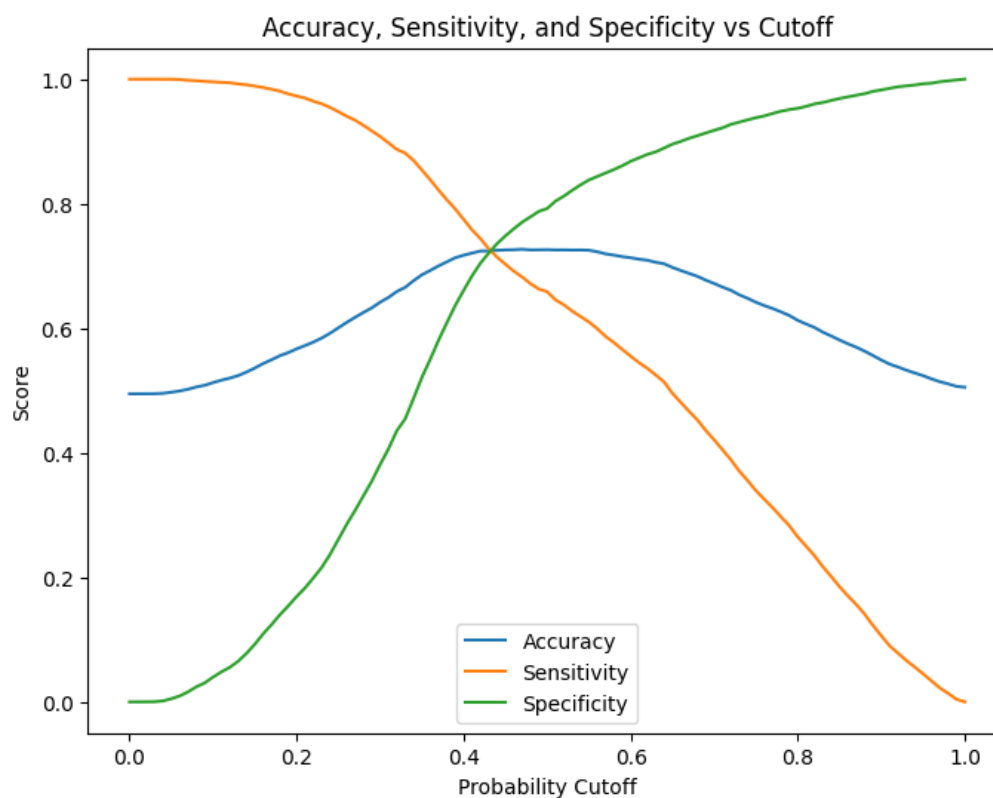


Figure 19: Accuracy, Sensitivity, and Specificity vs Cutoff for Linear SVM

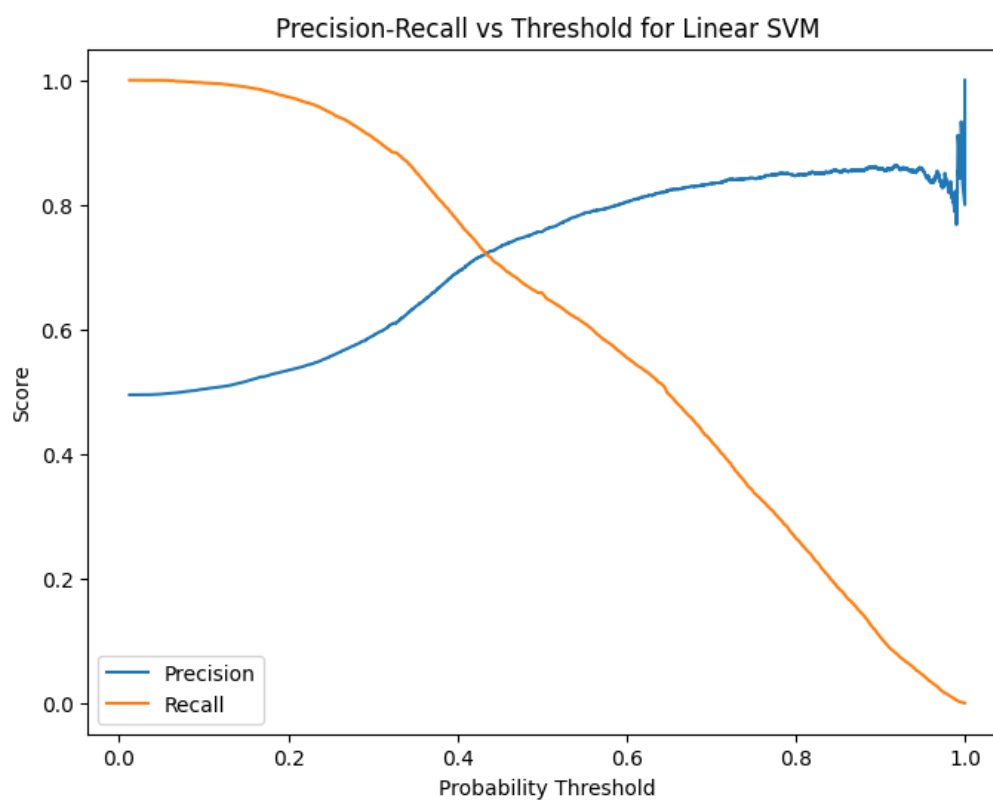


Figure 20: Precision-Recall vs Threshold for Linear SVM

C.2 Decision Tree

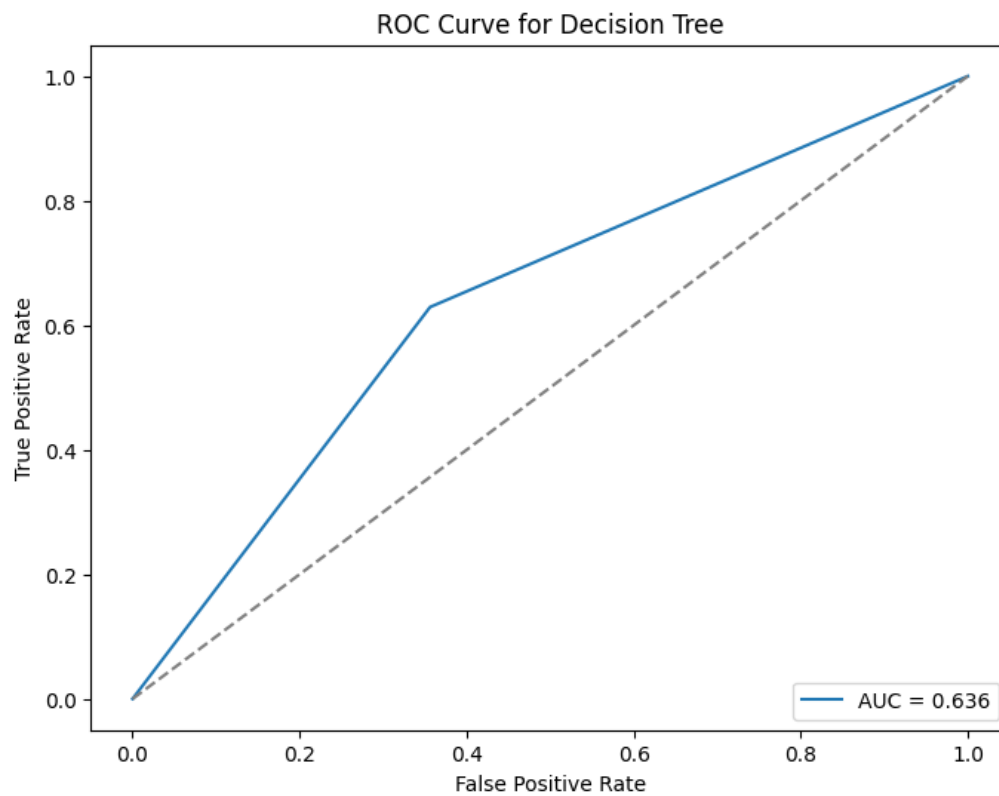


Figure 21: ROC Curve for Decision Tree

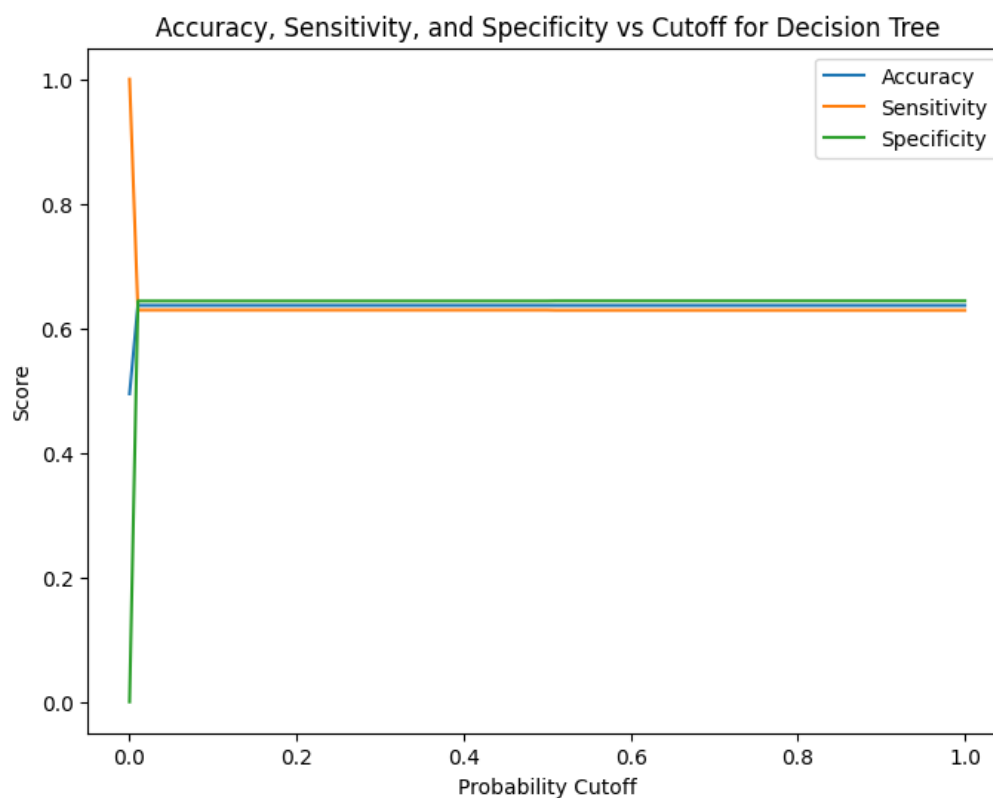


Figure 22: Accuracy, Sensitivity, and Specificity vs Cutoff for Decision Tree

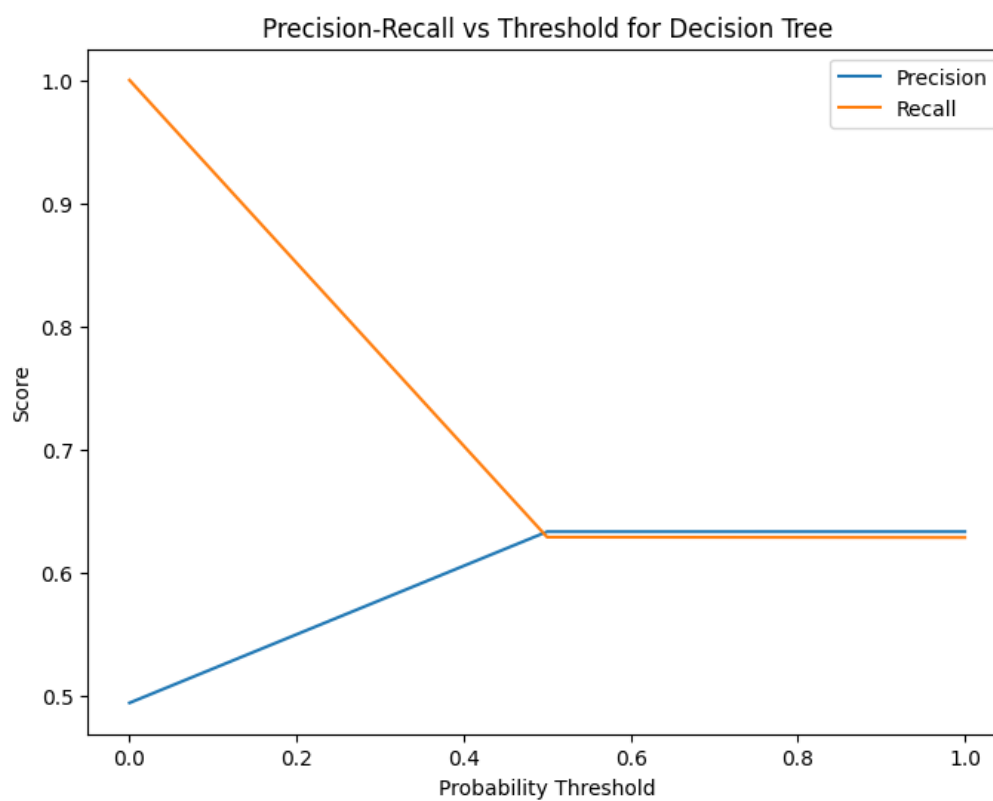


Figure 23: Precision-Recall vs Threshold for Decision Tree